STA4001 2021 Fall midterm WANG Youmo

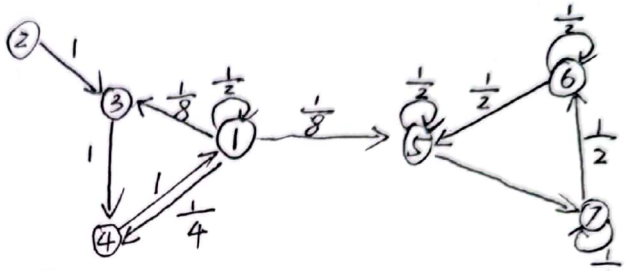
1. Solution. Let D be the event that a random person has this disease, and let P represent the event that the test for this person turns out positive. From the description of this question, we know that $P(D)=0.005$, $P(P|D^c)=0.03$ and $P(P^c|D)=0.02$. (a) $P(P)=P(P\cap D)+P(P\cap D^c)=P(P|D)\cdot P(D)+P(P|D^c)\cdot P(D^c)$ where $P(P|D)=1-P(P^c|D)=0.98$ $P(D^c)=1-P(D)=0.995$. Therefore,

$$P(P)=0.98\times 0.005+0.03\times 0.995=0.0049+0.02985=0.03475.$$

(b) $P(D|P)=\dfrac{P(D\cap P)}{P(P)}=\dfrac{0.98\times 0.005}{0.03475}=\dfrac{49}{347.5}\approx 0.141.$

2. Solution

$$P=\begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{8} & \frac{1}{4} & \frac{1}{8} & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$



(a) From the transition graph, we can see that.

① State 2 is not accessible from any other states, so $\{2\}$ is a class.

② State 1, 3, 4 communicate with each other. They are not accessible from state 4, 5, 6; and they cannot reach state 2. So $\{1,3,4\}$ is a class.

③ State 5, 6, 7 communicate with each other. They cannot reach any other states. Hence $\{5,6,7\}$ is a class. In summary, there are 3 classes: $\{2\}$, $\{1,3,4\}$, $\{5,6,7\}$.

(b) The class $\{2\}$ is transient as $P(X_n=2|X_0=2)=0$ for all n.

The class $\{1,3,4\}$ is also transient. Suppose $X_0=1$ and $X_1=5$, then this Markov chain never goes back to state 1. Thus, $1-f_{11}=P(X_n\neq 1 \text{ for all } n|X_0=1)\geq P(X_1=5|X_0=0)=\frac{1}{8}$. So $0<f_{11}\leq\frac{7}{8}<1$. As a result, class $\{1,3,4\}$ is transient.

Class $\{5,6,7\}$ is recurrent. Since this Markov chain only has 7 states, not all states can be transient. Since class $\{2\}$, $\{1,3,4\}$ are transient, $\{5,6,7\}$ must be recurrent.

3. Solution. We can construct a Markov chain $\{X_n\}$ with 4 states, and $X_n\in\{1,2,3,4\}$, which denotes that a product is damaged and will be discarded at Step 1, Step 2, and is made respectively. Then state 3 and 4 are two absorbing states, and the corresponding transition matrix is (a).

$$P=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.1 & 0.2 & 0.7 & 0 \\ 0.05 & 0.05 & 0.1 & 0.8 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(b). Let P_i be the probability that starting from state i, the DTMC is absorbed into state 0. Then our objective is to find out P_2, and we have

$$\begin{cases} P_1=1 \\ P_2=0.1P_1+0.2P_2+0.7P_3 \\ P_3=0.05P_1+0.05P_2+0.1P_3+0.8P_4 \\ P_4=0 \end{cases} \Rightarrow \begin{cases} P_1=1 \\ P_2=\frac{25}{137} \\ P_3=\frac{8}{137} \\ P_4=0 \end{cases}$$

Solving this system of linear equations, we get $P_2=\frac{25}{137}$.

4. Solution. A_n, B_n are the number of arrivals and departures at time n respectively.

(a) $A_n \sim$ Bernoulli(λ) $B_n \sim$ Geometric(μ) : the customer in service at each time will depart with probability μ independently. Hence, given $X_n \geq 1$, B_n also follows the Bernoulli distribution with parameter μ. However, given $X_n = 0$, it's reasonable that $B_n = 0$. $X_{n+1} = X_n + A_n - B_n$.

Verify the Markov Property: $P(X_{n+1}=j \mid X_n=i, X_{n-1}=i_{n-1}, \dots X_0=i_0) = P(X_n + A_n - B_n = j \mid X_n=i, X_{n-1}=i_{n-1}, \dots) = P(X_n - B_n + A_n = j \mid X_n = i) = P(X_{n+1}=j \mid X_n=i)$. Last equation holds since A_n doesn't rely on anything and B_n is only dependent on X_n. Now try to derive the transition probability:

① $X_n = 0$, $P_{0,j} = P(X_{n+1}=j \mid X_n=0) = P(X_n + A_n - B_n = j \mid X_n = 0) = P(A_n = j)$. Hence $P_{0,0} = P(A_n=0) = 1-\lambda$

$P_{0,1} = P(A_n=1) = \lambda$, and $P_{0,j} = 0$, for all $j \geq 2$.

② $X_n = i \geq 1$, we have $P_{i,j} = P(A_n - B_n = j - i \mid X_n = i)$. Hence $P_{i,i-1} = P(A_n=0) \cdot P(B_n=1 \mid X_n=i) \overset{\uparrow}{=} (1-\lambda)\mu$.

$P_{i,i+1} = P(A_n=1) P(B_n=0 \mid X_n=i) = \lambda(1-\mu)$. Therefore $P_{i,i} = \lambda\mu + (1-\lambda)(1-\mu)$ and $P_{i,j} = 0$, $\forall |i-j| \geq 2$.

Note that this means the DTMC can only make transitions from a state to itself/or one of its two nearest neighbors

(b). In order to prove the time reversibility, use the proposition

$$\begin{cases} \pi_i P_{ij} = \pi_j P_{ji} \\ \sum_i \pi_i = 1 \end{cases}$$

We derive $\pi_i = \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1} \pi_1 = \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1} \frac{\lambda}{\mu(1-\lambda)} \pi_0$. Since $\sum_i \pi_i = 1$,

$\pi_0 = \left[1 + \sum_{i=1}^{\infty} \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right]^{-1}$ Plug $\lambda = 0.6$, $\mu = 0.9$ into the equation $\pi_0 = \left[1 + \frac{5}{3}\sum_{i=1}^{\infty} \left(\frac{1}{6}\right)^{i-1}\right]^{-1} = \frac{1}{3}$. and for any state $i \geq 1$,

$\pi_i = \frac{10}{3} \times \frac{1}{6^i}$ Since π is a distribution, then it is stationary and the Chain is time reversible

(c). $\begin{cases} \pi_j = \sum_i \pi_i P_{ij} \\ \sum_i \pi_i = 1. \end{cases} \Rightarrow \pi_i = \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1} \frac{\lambda}{\mu(1-\lambda)} \pi_0$. Similarly, $\pi_0 = \left\{1 + \sum_{i=1}^{\infty} \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1} \frac{\lambda}{\mu(1-\lambda)}\right\}^{-1}$

Plug $\lambda = 0.9$, $\mu = 0.6$ into the equation, $\pi_i = 0$, for all $i \geq 0$. So X_n does not have a stationary distribution

(d) X_n will go to infinity with probability 1 as $n \to \infty$

(e) If $\mu > \lambda$, then $1-\mu < 1-\lambda$ and hence $\frac{\lambda(1-\mu)}{\mu(1-\lambda)} < 1$. Thus, $\sum_{i=1}^{\infty} \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1}$ converges and we can get stationary distribution from (b). Otherwise, if $\mu \leq \lambda$, then $\sum_{i=1}^{\infty} \left[\frac{\lambda(1-\mu)}{\mu(1-\lambda)}\right]^{i-1}$ will go to infinity. From (c), the stationary probability $\pi_i = 0$ for all $i \geq 0$. so X_n does not have a stationary distribution.

(f) $\lambda < \mu_1$ and $\lambda < \mu_2$.